MSc thesis in Geomatics

Deep Learning Classification Leveraging Multi-Temporal Historical Maps for Enabling Urban Analysis in The Past

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Andika Hadi Hutama: *Deep Learning Classification Leveraging Multi-Temporal Historical Maps for Enabling Urban Analysis in The Past* (2026)

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Abstract

Info

Should fit on one page.

Lemongrass frosted gingerbread bites banana bread orange crumbled lentils sweet potato black bean burrito green pepper springtime strawberry ginger lemongrass agave green tea smoky maple tempeh glaze enchiladas couscous. Cranberry spritzer Malaysian cinnamon pineapple salsa apples spring cherry bomb bananas blueberry pops scotch bonnet pepper spiced pumpkin chili lime eating together kale blood orange smash arugula salad. Bento box roasted peanuts pasta Sicilian pistachio pesto lavender lemonade elderberry Southern Italian citrusy mint lime taco salsa lentils walnut pesto tart quinoa flatbread sweet potato grenadillo.

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Acknowledgements

Thanks to everyone, especially my supervisors and my dog (in that order). Obviously, thanks also to the brilliant minds who created this *fantastic* template!

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Thai super chili apricot salad cocoa dark chocolate vitamin glow mushroom risotto red amazon pepper simmer udon noodles soba noodles dragon fruit cherries strawberry mango smoothie basil chickpea crust pizza cauliflower cherry bomb pepper mediterranean street style Thai basil tacos. Balsamic vinaigrette Indian spiced kimchi tofu sandwiches smoked tofu apple vinaigrette salty Thai sun pepper cayenne four-layer fiery fruit peach strawberry mango vegan Bulgarian carrot Italian linguine puttanesca green bowl lemon red lentil soup overflowing berries habanero golden one bowl.

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TODOs

1. Introduction

This chapter begins with an introduction to the problem statement and motivation behind this research, followed by the research objectives and questions. The scope of the research is defined. Ultimately, an outline of the thesis structure is provided to guide the reader through the subsequent chapters.

1.1. Introduction

Historical maps are widely recognized as valuable resources for understanding the transformation of geographical space over time, particularly in the fields of historical and urban studies. The rapid development of geodesy and cartography from the 18th century resulted in extensive production of topographic maps at various scales across the Western world. Among these, urban maps are especially informative due to their high level of detail and often reliable geometric representation of spatial features (Chen et al., 2021). By integrating these maps within modern geospatial systems, researchers can gain valuable insights into the processes that shaped present-day urban configurations (Nobajas & Nadal, 2015).

According to urban studies conducted in the Netherlands (Rutte, 2023, p. 14) the underlying spatial structures of towns and cities such as street patterns, parcel layouts, and drainage networks, tend to persist over centuries, a phenomenon described as "inertia", implying that present-day urban configurations cannot be properly understood without tracing their origins through historical map series. Such studies were made possible by the extensiveavailability of historical map collections in the Netherlands, such as the *Topografische en Militaire Kaart van het Koningrijk der Nederlanden* (TMK) series (1850-1865), the *Chromotopograische kaart des Rijks* series (1884-1930), and numerous municipality-specific historical map series.

However, despite the rich availability of historical map collections and recent digitization efforts that have made large map collections browsable on screen, their contents remain largely inaccessible for computational analysis, because they are not yet machine-readable (Hosseini et al., 2022). Extracting geospatial information from these maps presents several challenges, including the heterogeneity and inconsistency in cartographic styles across different map series, the visual complexity introduced by overlapping text, symbols, and digitization artifacts, as well as the degradation of map quality over time (Chen et al., 2021). Furthermore, before any computational analysis can be performed, these maps must first be accurately georeferenced, a prerequisite step that is particularly demanding for maps with low planimetric accuracy, often requiring substantial manual effort (Vaienti et al., 2025). Moreover, the time-consuming nature of converting maps into machine-readable formats remained unfeasible over large geographical areas (Goodchild, 2018). The scalability limitation has historically constrained historical map analysis to case studies with smaller size of datasets, preventing computational analysis of larger areas needed to address complex spatiotemporal questions (Liu et al., 2024).

In recent years, computer-vision-based deep learning has emerged as a promising approach for automating feature extraction from satellite imagery and has been increasingly applied to historical map analysis, demonstrating good performance in dectecting urban features, road networks, and land-cover classes (O'Hara et al., 2024), (Xia et al., 2025). This shift has opened new possibilities for processing historical maps at a scale previously unfeasible through manual or semi-automated workflows (Hosseini et al., 2022).

However, deep learning methods are notoriously data-hungry, requiring large volume and high-quality annotated training data to achieve reliable performance. Producing such annotations is a costly and labor-intensive process, as each map sheet must be manually interpreted and labelled, a task that contradicts the very objective of automation (Arzoumanidis et al., 2025). This challenge is further amplified by the stylistic variability across different map series where different cartographic conventions, color palettes, and map quality across time periods and map series limit the ability of models trained on one collection to generalize to another (Arzoumanidis et al., 2025).

Automated extraction of urban features from historical maps can be approached at different levels of granularity. At the broader level, patch-based methods classify map regions to delineate urban extent, enabling statistical-level analysis or urban growth across large geographic areas. At a finer-level, object-based methods aim to detect and delineate individual objects, particularly building objects, offering more detailed spatial information but at greater computational cost and data demand. Each approach carries distinct trade-offs in terms of spatial detail, scalability, and processing requirement.

1.2. Research Objectives

Addressing the challenges and trade-offs in the previous Section 1.1 is essential for enabling reproducible urban change analysis from historical map series. Therefore, this thesis investigates and compares patch-level and object-level deep learning approaches for automated urban extent extraction from Dutch historical topographic maps, with the aim of assessing their respective effectiveness for enabling urban change analysis across multiple temporal periods.

The main research question of this thesis is: *"To what extent do patch-level and object-level deep learning approaches differ in their effectiveness for extracting urban extent from multi-temporal historical topographic maps for enabling urban change analysis?"*

To address the main question, the following sub-research questions are as follows:

1. "How can patch-level deep learning classification be applied to extract urban extent from multi-temporal historical maps?"
2. "How can CycleGAN-based synthetic data generation be applied to enable object-level building footprint extraction from historical maps?"
3. "What urban change patterns can be identified by comparing automatically extracted building extents from historical maps against reference data?"
4. "What are the relative strengths, limitations, and trade-offs between patch-level and object-level deep learning approaches for urban extent from historical maps?"

1.3. Scope

The following clarifies the scope of this thesis:

- This study only uses historical maps that are already scanned, and digitally available to access online from authoritative sources. Besides this study only uses historical maps that are already georeferenced.
- The use of additional historical data sources (cadastral records, building registries, old photographs, paintings, or textual archives) is not included in this study.
- Urban extent in this study refers to the horizontal built-up footprint as depicted on the map surface, such as building blocks and individual building footprints. Vertical urban change such as building height or densification, and administration-based boundary definitions are not considered.

1.4. Thesis Outline

This thesis is structured into six main chapters, the outline of which is as follows:

Chapter 1 gives the overview of the research, including the background and motivation, followed by research objectives, research questions, scope of research, and lastly, the outline to the thesis.

Chapter 2 provides an overview of scientific related work and theoretical background related to the thesis topic. It begins with a review of previous studies on historical maps for urban studies. Followed by the use of IIIF in supporting access of historical map collections, existing methods of feature extraction from historical maps, and deep learning approaches.

Chapter 3 provides the detailed description of the research methodology, including the approaches for data processing, model training, evaluation, and post-processing steps.

Chapter 4 describes the datasets and study area used in this thesis. Followed by the description of programming tools, software, and hardware implemented to test the methodology.

Chapter 5 presents the result and findings of the experiments and analyses conducted in this thesis.

Chapter 6 discusses the main findings of the research, addresses the research questions, and provides recommendations for future work.

2. Related Work

In this chapter, an overview of the related work relevant to the thesis is provided. First section discusses the use of historical maps as sources for urban studies (Section 2.1). Then, the development and contribution of IIIF and Allmaps is given (Section 2.2). Lastly, existing feature extraction method from historical maps (Section 2.3), and relevant deep learning methods are described (Section 2.4).

2.1. Historical maps as sources for urban studies

The Netherlands has a particularly rich tradition of historical cartographic documentation, offering one of the most extensive national map series in Europe. The *Kraijenhoff* map (1798-1822), *Topographische en Militaire Kaart van het Koningrijk der Nederlanden* (TMK, 1850-1865) and the *Chromotopografische Kaart des Rijks* (Bonnebladen, 1865-1949) together provide near-complete topographic coverage of the Netherlands across multiple decades, offering a unique multi-temporal record of the Dutch landscape. These series are further complemented by municipality-specific and thematic historical map series such as *Waterstaatkaart*, which offer considerably higher level of cartographic detail for specific regions and topics. The value of these collections for urban studies is well demonstrated by the *Historische Atlas van Nederland* (Rutte, 2023), which traces the spatial evolution of Dutch towns and cities from the medieval period to the twentieth century, documenting the dramatic expansion of built-up areas during the industrial era. More recently, institutional efforts have sought to make such collections digitally accessible. The Dutch Kadaster launched *Topotijdreis* (Kadaster, 2015)[1], an online platform providing access to over 200 years of topographic maps (illustrated in Figure 2.1), allowing users to explore Dutch landscape and how it has evolved over time. Despite these efforts, these digitized map collections remain in raster image format and are not yet machine-readable, limiting their use to visual inspection only and preventing direct computational querying or geospatial analysis.

Nevertheless, where such processing has been carried out, typically through manual digitization or automated workflows, historical map series have demonstrated considerable value for studying long-term urban growth and land use change at multiple scales. The *Historisch Grondgebruk Nederland* (HGN) project produced a nationwide digital land use dataset for the Netherlands around 1900, derived from historical topographic maps, providing one of the earliest spatially explicit records of Dutch land use before the remote sensing era (Knol et al., 2004). The HisGIS 1832 project digitized and georeferenced the first systematic cadastral survey of the Netherlands, producing a nationwide spatial database of land ownership and building footprints from 1832 (Humanities Cluster of the Royal Netherlands Academy of Arts and Sciences (KNAW), 2024). At the city level, studies such as those synthesized in the *Historische Atlas van Nederland* (Rutte, 2023) have quantified built-up area expansion for major Dutch cities between 1850 and 1900, revealing growth rates of several hundred percent in industrializing cities such as Rotterdam and Apeldoorn. These studies demonstrate that multi-temporal historical map series are capable of supporting statistical-level urban change analysis: measuring aggregate shifts in built-up extent, land use composition, and spatial distribution of urban development. However, such studies have largely been constrained to manually digitized datasets covering limited geographic extents, as the absence of scalable automated extraction methods has made nation-wide computational analysis infeasible.

[1] Topotijdreis website: https://topotijdreis.nl/

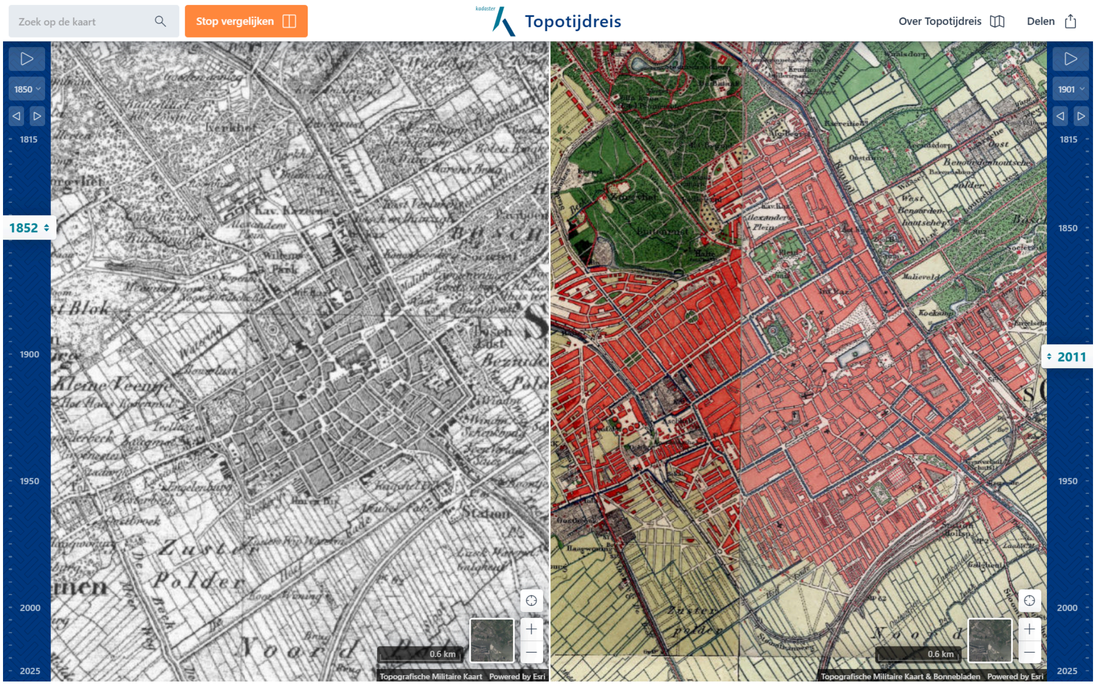

Figure 2.1: The web application of *Topotijdreis* developed by the Kadaster

2.2. International Image Interoperability Framework (IIIF) and Allmaps

A prerequisite for any large-scale computational analysis of historical maps is reliable programmatic access to digitized map collections, which are typically stored as high-resolution image files distributed across multiple institutional repositories. The International Image Interoperability Framework (IIIF) addresses this challenge by defining a set of open standards for delivering digital image objects online at scale, enabling consistent programmatic access through standardized APIs regardless of the hosting institution (Snydman et al., 2015). Cultural heritage institutions in the Netherlands, including the Dutch National Archives, have adopted IIIF to serve their digitized map collections, making thousands of historical map sheets retrievable through uniform URL-based requests.

Building on IIIF, the Allmaps[2] platform extends this infrastructure to specifically for cartographic collections, providing tools to curate, georeference, and explore digitized maps from any IIIF-compliant repository without requiring complex GIS infrastructure (Hanlon et al., 2025). All georeferencing information is stored in a standardized georeference annotation, an approved extension to the IIIF Presentation API based on the W3C Web Annotation standard (IIIF Maps Community Group, 2023). A georeference annotation records the Ground Control Points (GCPs) that map pixel coordinates of the historical map image to real-world geographic coordinates, along with an optional transformation algorithm (e.g. Thin Plate Spline, Polynomial). The resulting annotation can be used to warp the map into a standard Coordinate Reference System (CRS), enabling direct client-side visualization through web mapping applications such as XYZ tile services, as well as integration into GIS tools such as QGIS for geospatial analysis workflows.

While IIIF and Allmaps resolve the distribution and access challenge, they do not address the semantic extraction problem. Maps served through IIIF remain raster images without machine-

[2] Allmaps website: https://allmaps.org/

readable content, and extracting meaningful geospatial features from them still requires additional processing methods, which is the focus of the following sections.

2.3. Feature extraction from historical maps

Before historical maps can be used in any quantitative spatial analysis, they must undergo a multi-step processing chain to transform scanned raster images into machine-readable geodata. This chain typically involves digitization, georeferencing, image classification or segmentation, and post-processing to filter unwanted feature as text and symbols (Gobbi et al., 2019). The output of this process can take two forms: raster data, where each pixel is assigned a thematic class, or vector data, where spatial features are represented as discrete geometric objects such as polygons and lines. Raster outputs are well-suited for large-scale spatial analysis and land use change studies, while vector outputs enable more precise object-level queries and integration with modern GIS (Geographic Information System) datasets (Gobbi et al., 2019). Several main approaches of digitization map can be identified into four categories: manual, semi-automatic, machine learning and deep learning classification.

2.3.1. Manual digitization

The extraction of spatial information from historical maps has traditionally relied on manual digitization since the emergence of GIS technology. In this approach, trained operators visually interpret the map and manually trace geographic features such as building footprints, road networks, and land parcel boundaries, assigning class labels or attribute values to each delineated object (Gobbi et al., 2019). The approach is valued in digital humanities for the interpretive depth it enables: operators can resolve ambiguous features, apply contextual knowledge, and produce geometrically precise vector outputs that automated methods often struggle to match (Chen et al., 2021).

A prominent example in the Dutch context is the HisGIS project[3], which manually digitized and vectorized the Napoleonic cadastral survey of 1832, producing a nationwide spatial database of approximately 3 million land parcels covering the entire Netherlands, with attributes including parcel size, owner and land use for each units (Humanities Cluster of the Royal Netherlands Academy of Arts and Sciences (KNAW), 2024). The scale and detail of HisGIS illustrates what manual digitization can achieve when organized as a sustained, multi-year collaborative effort. Collaborative approaches and large-scale crowdsourcing experiments have also been proposed to handle large map collections (Budig et al., 2016; Southall et al., 2017), distributing the annotation workload across many contributors. However, even with crowdsourcing, manual processes remain tedious, time-consuming, and non-reproducible (Chen et al., 2021).

2.3.2. Semi-automated classification

To reduce the reliance on manual labor, semi-automated methods have been developed that use image processing algorithms to automatically classify map pixels or regions, while still requiring human intervention to define parameters or provide reference samples. The main techniques developed for this purpose include histogram thresholding, color space clustering, edge detection, region growing, and Object-Based Image Analysis (OBIA) (Chiang et al., 2014; Gobbi et al., 2019). These methods exploit the cartographic properties of printed maps, particularly the use of distinct colors, line patterns, and spatial boundaries to encode different feature types, to separate target classes.

[3] HisGIS project website: https://hisgis.nl/

A notable application in the Dutch context is the *Historisch Grondgebruk Nederland 1900* (HGN1900) project (Knol et al., 2004), which produced a nationwide digital land use dataset for the Netherlands around 1900, derived from the historical topographic maps using color-based image classification and manual post-correction. While the result achieved an overall classification accuracy of 95.9% across 6,300 national validation points (Knol et al., 2004), a fundamental limitation of its color-based approach is the inability to distinguish between features, buildings and paved roads are both represented in red on the Bonne maps, resulting in a single merged class that conflates two distinct urban features.

Gobbi et al. (2019) proposed an OBIA-based workflow for semi-automatic digitization of heterogeneous map collections, combining region-based image segmentation with supervised machine learning classifiers and an automated text and symbol removal module. Applied to three maps with very different characteristics, this approach achieved average classification of 97%, marking a significant improvement over manual methods.

Despite these advances, semi-automated methods carry important limitations. The correct parameters such as color thresholds, segmentation scales, or classifier training samples must be determined and tuned individually for each map or map series, as differences in printing quality, paper degradation, and cartographic style across map editions can significantly affect classification performance (Chen et al., 2021)

2.3.3. Machine learning and deep learning classification

The limitations of rule-based semi automated method have motivated the adoption of machine learning and, more recently, deep learning approaches, which learn feature representations directly from annotated training data rather than relying on manual defined rules. Compared to traditional machine learning classifiers such as Random Forest or Support Vector Machines, which require hand-crafted feature descriptors as input, deep learning models, particularly Convolutional Neural Network (CNN), can automatically learn hierarchical visual features from raw image patches, offering greater flexibility and robustness across varying cartographic styles (Heitzler & Hurni, 2020; Hosseini et al., 2022).

CNN-based methods have been increasingly applied within the historical maps domain. At the patch level, image classification approaches assign a class label to each image patch, and have been used to detect urban settlements and built-up areas from historical topographical map sheets (Hosseini et al., 2022; Uhl et al., 2017). At a finer level of detail, semantic segmentation models, most notably U-Net (Ronneberger et al., 2015), assign a class label to every pixel in the image and have been successfully applied to extract building footprints (Heitzler & Hurni, 2020), road networks, and wetland symbology (O'Hara et al., 2024). For tasks requiring individual object delineation, instance segmentation such as Mask R-CNN produce a separate mask for each detected object and have been applied to text detection and segmentation in historical maps, supporting downstream text removal and vectorization workflows (Suresh Dodda, 2024). These four task types, namely image classification, object detection, semantic segmentation, and instance segmentation, represent the main computer vision approaches used for historical map content extraction and form the methodological basis for the extraction approaches investigated in this thesis.

While these approaches have demonstrated promising results, their application to historical maps introduces a set of specific challenges that are discussed in the following section.

2.4. Deep learning for historical map analysis

Despite the promising results demonstrated by CNN-based approaches, deep learning methods share a common and fundamental requirement: large volumes of annotated training data. Producing such annotations is a costly and labor-intensive process, as each map sheet must be manually interpreted and labeled, which contradicts the very goal of automation (Arzoumanidis et al., 2025). ThiAs challenge is particularly significant for models developed for a specific map series or historical period, where the amount of available training data is limited in both geographic coverage and time span.

The problem of data scarcity is further compounded by the diversity of historical map styles. Models trained on a homogeneous corpus, such as a single national map series, tend to perform well within that corpus but struggle to generalize to maps with different cartographic styles, color palettes, or symbology (Petitpierre et al., 2021). Petitpierre et al. (2021) confirmed this through experiments across two diverse map corpora, showing that stylistic variability significantly degrades segmentation performance, and that developing a truly generic model for heterogeneous map collections remain an open challenge.

To address training data scarcity, image-to-image translation methods have emerged as a promising strategy for generating synthetic training data. Zhu et al. (2020) demonstrated image-to-image translation applied to several visual domain transfer tasks, including the translation of aerial photographs to Google Maps style and vice versa. In their work, they distinguished between two translation paradigms: paired translation using pix2pix framework, which requires matches input-output image pairs, and unpaired translation using CycleGAN, which learns to transfer styles between two domains without requiring any direct correspondence between images. For historical maps, the unpaired approach is particularly relevant, as paired correspondences between a historical map and its equivalent ground truth rarely exist. This approach offers a practical path toward reducing the annotation burden for historical map analysis and forms the basis for synthetic training data strategy adopted in this thesis.

A related limitation concerns the use of large pre-trained foundation models. Models such as the Segment Anything Model (SAM) and Vision Transformer (ViT)-based architectures have shown strong performance across general computer vision tasks, but their training on natural images creates a domain mismatch when applied to historical maps, leading to poor zero-shot performance on cartographic content (Xia et al., 2025). Xia et al. (2025) addressed this by adapting SAM specifically for historical map feature detection through domain-specific fine-tuning, demonstrating that generic foundation models require substantial adaptation before they can be reliably applied to historical maps.

Beyond the training data challenge, deep learning pipelines for historical map analysis, typically require multi-step post-processing workflows to convert pixel-level model outputs into usable vector geometries (Heitzler & Hurni, 2020). Each additional step in this pipeline introduces opportunities for error propagation, and the engineering effort required to design and maintain these steps for each specific map type reduce the overall degree of automation (Xia et al., 2024). This limitation underlines the need for extraction approaches that are not only accurate, but also practical and transferable across different map editions without requiring extensive map-specific tuning.

3. Methodology

This chapter describes the methodology developed to automatically extract urban extent from historical topographic maps. In general, this thesis is divided into 2 main methodologies: (1) patch-based classification method and (2) object-based classification method. The first method is based on a patch-based classification approach, where the map is divided into small patches and each patch is classified as urban or non-urban using a deep learning model, from this point afterwards in this report will be referred as METHOD 1. The second method is based on an object-based classification approach, where individual building footprints are extracted as objects alongside it's geometry, which will be referred as METHOD 2. Both methods are implemented using deep learning techniques, specifically convolutional neural networks (CNN) based architectures. The methodology also includes a change analysis step to analyze the urban transformation across multiple temporal periods. The main difference of the 2 methodologies lies in their granularity of target object, where one focuses on patches and the other on individual buildings. Figure 3.1 and Figure 3.2 displays the general methodology workflow for each method. A complete version of implemented steps is given in Appendix.

insert complete workflow

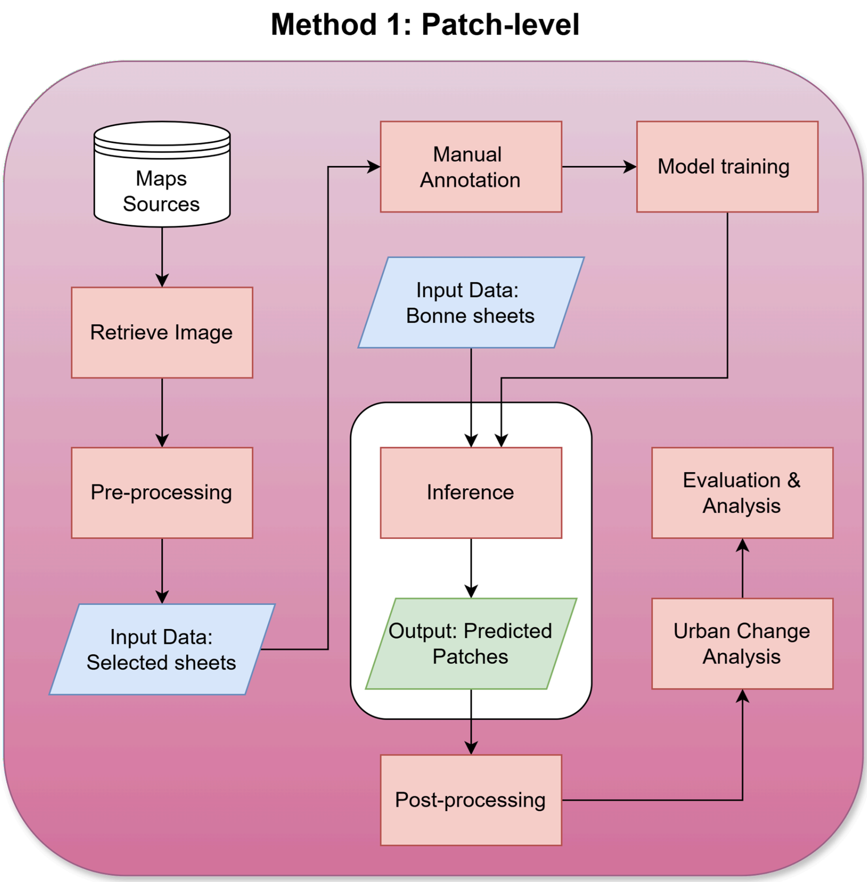

Figure 3.1: (a) Method 1: Patch-level

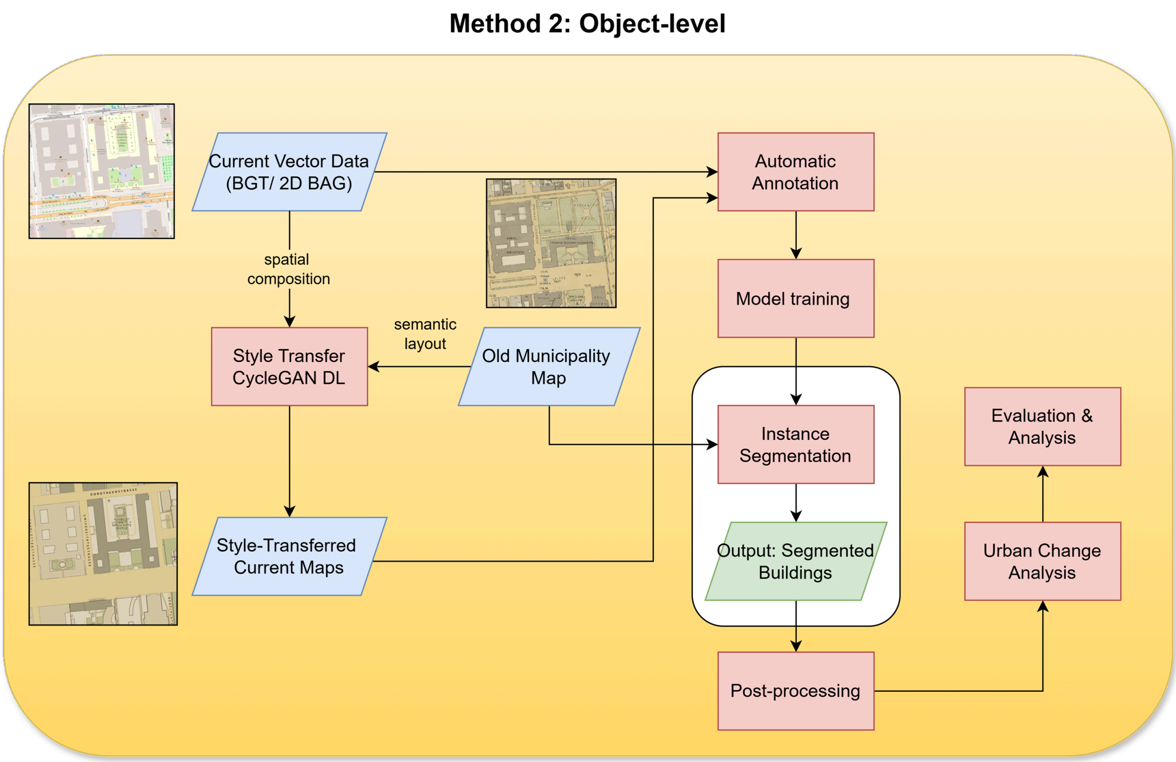

Figure 3.2: (**b**) Method 2: Object-level

The following sections provide a detailed description of each methodology.

3.1. Method 1: Patch-level classification

In the following sections, the complete pipeline from data acquisition until post-processing is described.

3.1.1. Data acquisition and preparation

3.1.1.1. Map projection

3.1.2. Manual annotation

3.1.3. Model training

3.1.4. Inference

3.1.5. Post-processing

3.1.6. Evaluation

TODO: Hints

1. Selesaiin Method 2 dulu (karena dikerjain terakhir)
2. Baru Method 1 (sambil inget2 lagi nanti)

3.2. Method 2: Object-level classification

Method 2 extracts individual building footprints as vector polygons from the Rotterdam 1897 municipality map series using a four-stage pipeline: (1) data acquisition and preparation, (2) synthetic training data generation via CycleGAN, (3) instance segmentation via Mask R-CNN, and (4) post-processing.

3.2.1. Data acquisition and preparation

TODO: Key pointers

- Explored multiple sources of historical maps which comply to the requirement criteria, selected: Rotterdam 1897 map sheets.
- Download the map sheet using Allmaps CLI, export to GeoTIFF
- Download BGT using QGIS Plugin BGT Downloader
- List relevant BGT layers for cartographic style adjustment
- Allmaps XYZ Tile server to help visualization
- Inspect all layers in QGIS

Method 2 uses two primary datasets: the Rotterdam 1897 municipality maps and the current BGT vector data, both described in Section 4.1.

The historical map sheets are downloaded using the Allmaps CLI, which exports each sheet as a georeferenced GeoTIFF using the georeference annotations available on the Allmaps platform (Section 3.2.2). Of the 10 sheets in the series, only sheets 2, 3, and 4 are used, as these cover areas with dense building content. The remaining sheets are dominated by water bodies and open land.

BGT data are downloaded via the BGT Downloader plugin in QGIS. The relevant layers selected for this research are:

1. *Pand* (buildings)
2. *Wegdeel* (road surfaces)
3. *Waterdeel* (water bodies)
4. *BegroeidTerreindeel* (vegetated areas)
5. *OnbegroeidTerreindeel* (bare surfaces)

While the primary segmentation target is buildings, the background layers are included because they form the visual context that looks similar to the historical map sheets. Then, all downloaded data are inspected in QGIS to verify spatial coverage, alignment with the historical map extent, and layer completeness. The Allmaps XYZ Tile server is additionally used as an alternative to directly fetch IIIF resources as reference background in QGIS, without any preprocessing steps.

3.2.2. Georeference

TODO: Key pointers

- Manual georeference

The georeference annotations for the Rotterdam 1897 sheets, maintained by the TU Delft Library and Allmaps team, are applied using the Allmaps CLI to produce georeferenced GeoTIFFs in EPSG:3857. A qualitative assessment of georeferencing quality is discussed in Section 5

complete this section

3.2.3. Synthetic data generation (CycleGAN)

TODO: Key pointers

- Vector style transfer: configure cartographic style of current building vector data (BGT)
  - Visual inspection -> compute mean RGB from selected patches using 3x3 kernal
- Preprocessing -> `tile_patches.py`, verify (remove invalid patches), and data splitting into trainA/B and testA/B for CycleGAN input
- CycleGAN for image translation
  - Conceptual of CycleGAN (brief theory, figures from the paper, formula, loss,)
  - Training the model (`train.py`)
  - Testing (`test.py`)
  - Evaluation (Compute FID, loss metrics from training, visual observation)
  - Output: `model.pth` cycleGAN patches (PNGs)
  - Post processing - reconstruct (`stitch_fake_b.py`) -> GeoTIFF for input for mask r-cnn

To generate annotated training data without manual labelling, this research adopts a synthetic data bootstrapping approach following Arzoumanidis et al. (2025). The pipeline consists of two steps: (1) cartographic style transfer, which renders the BGT vector data to visually resemble the historical maps, and (2) CycleGAN image translation, which adds realistic cartographic degradation effecs to the style-transferred images. The BGT vector geometry simultaneously serves as perfectly aligned annotation masks, eliminating the need for manual ground truth annotation.

3.2.4. Model training (Mask R-CNN)

3.2.5. Inference

3.2.6. Post-processing

Building regularisation

3.2.7. Evaluation

3.3. Urban Change Analysis

(shared for both methods)

4. Implementation

This chapter begins by outlining all the datasets required for conducting this research, including the historical maps and supporting datasets as described in Section 4.1. Section 4.2 describes the study area for each method. In addition, the tools and hardware used to implement and test the methodologies are also described in Section 4.3 and Section 4.4 respectively.

4.1. Datasets

As mentioned in the previous chapter, this research involves 2 main methodologies, each method requires different datasets. For Method 1, multi-temporal Dutch historical topographic maps or the *Bonnebladen* map series is used as the main dataset. Subsequently, historical land use data (*Historisch Grondgebruik Nederland* - HGN1900) are used as reference data for evaluation.

For Method 2, historical municipality maps (i.e. *Gemeente Rotterdam Plattegrond 1897* and *Brandgrens kaart*) and current building footprint data from *Basisregistratie Grootschalige Topografie* (BGT) are used as the main datasets.

Additionally, several supporting datasets are also used for the implementation of both methods, such as current administrative boundaries and topographic base maps. All datasets used in this research originate from their respective sources and are available through open-access means. The following sections provide a detailed description of each dataset.

4.1.1. Bonnebladen map series

Bonnebladen maps is the chromo-topographical map of the Kingdom of the Netherlands on the scale of 1:25,000, which was originally produced by the former *Topograhisch Bureau* between 1865 to 1949 (Kadaster et al., 1940). The map series can be accessed from the Data Station Physical and Technical Sciences (DANS) repository in JPEG scans and GeoTIFF formats via their website (Link: https://phys-techsciences.datastations.nl/dataset.xhtml?persistentId=doi:10.17026/dans-xqf-r4xf). Relevant materials such as data catalogue, map legends, and metadata are also available on the website. To access the dataset, users are required to create a DANS account and request access to the dataset. Example of a single map sheet of Delft area is shown in the Figure 4.1 below.

Alternatively, the same datasets are also available in the form of IIIF resources, which are organized and maintained by the TU Delft Library. An interactive web viewer is established to facilitate the access and visualization of the dataset, which can be accessed via the following link: https://observablehq.com/d/ded63244d9f6625d.

Bonnebladen map series is selected as the main dataset for method 1 because of its rich historical map collection, which covers the entire Netherlands and spans a long temporal period, offers flexibility for defining the thesis scope. The consistent cartographic style with the addition of information of color is also beneficial for the implementation of computer-vision based deep learning techniques. More details about the dataset is discussed further in Section 5.

4.1.2. HGN1900 (Historisch Grondgebruik Nederland)

The *Historisch Grondgebruik Nederland* (HGN1900) dataset is a historical land use dataset that provides information about the land use in the Netherlands around the year 1900 (Knol et al., 2004). The dataset is available in raster format with resolution 50 meters per pixel and georef-

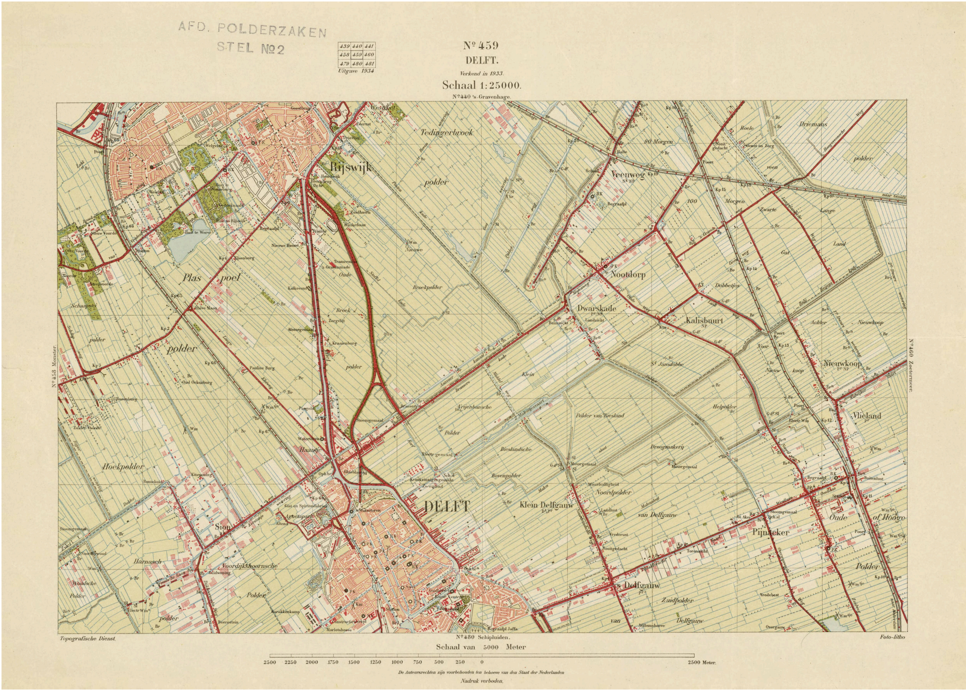

Figure 4.1: Single sheet of Bonnebladen - sheet no 449 - Delft

erenced with the RD coordinate system. The final product of the dataset is a land use map with 10 classes, which are derived from a semi-automatic classification from the original scanned Bonnebladen maps. The dataset is available for download from the website of the Land Use Database of The Netherlands (LGN) with the following link: https://lgn.nl/bestanden. The dataset is selected as reference data for method 1 because it provides a comprehensive land use map of the Netherlands around the same period as the Bonnebladen maps, which allows for evaluation of the results.

4.1.3. Historical municipality maps of Rotterdam

Map of Rotterdam (1897) sheets are the topographical maps series (1:5000 map scale) obtained from the Rotterdam City Archives. The series contains 10 sheets for the city of Rotterdam with published date around the year 1897.

The link to access the dataset is as follows: https://stadsarchief.rotterdam.nl/zoek-alles?mivast=184&mizig=299&miadt=184&miview=ldt&milang=nl&micode=4201&minr=39521873&miaet=14. Additionally, these map series has been processed and georeferenced by the TU Delft Library and AllMaps team, and made available as IIIF resources. A public interactive web viewer is established to facilitate the access and visualization of the dataset, which can be accessed via the following link: https://observablehq.com/@allmaps/rtm-historical-atlas.

Following this, another historical maps from the city archives, the *Brandgrens kaart* (*1940*) or fire boundary map with 1:5000 map scale, is also used as a supporting dataset for method 2. The map provides information about the extent of the fire that occurred in Rotterdam during World War II, particularly the bombing of May 14th 1940, which can be used to analyze the urban transformation before and after the war. Figure 4.2 and Figure 4.3 show the sample sheets of the Rotterdam municipality map and the fire boundary map respectively.

The dataset is selected as the main dataset for method 2 because of its detailed representation of the city of Rotterdam in the late 19th century, which allows for the extraction of individual

Figure 4.2: Sample sheet of Rotterdam municipality map - sheet no 3

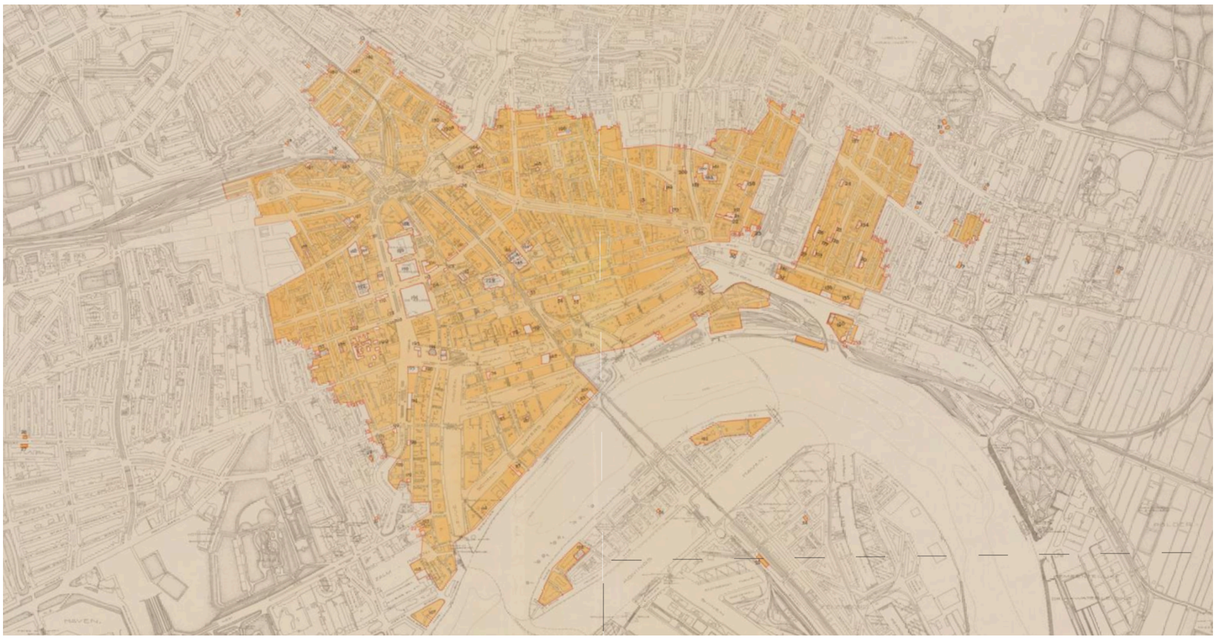

Figure 4.3: Fire boundary map of Rotterdam

building footprints as objects, instead of building blocks as in other historical maps with lower map scale.

4.1.4. Basisregistratie Grootschalige Topografie (BGT)

The *Basisregistratie Grootschalige Topografie* (BGT) is a large-scale topographic dataset that provides detailed information about the built environment in the Netherlands. The dataset includes information about buildings, roads, water bodies, and other topographic features. The building footprints in the BGT dataset are represented as polygons with relevant attributes to the research such as function and construction year. The dataset is available for download from the website of the Dutch Kadaster with the following link: https://api.pdok.nl/lv/bgt/ogc/v1/.

4.2. Study Area

For method 1, the research focuses on the Province of South Holland, which is one of the most urbanized regions in the Netherlands. While for method 2, the research limits the study area to the city of Rotterdam. Due to World War II, the city of Rotterdam has undergone significant urban transformation, which makes it an interesting case study for analyzing urban change over time. The choice of the study area is based on the availability of data as well as the relevance to the research topic, where both areas have rich historical map collections and have undergone significant urban transformation in the past.

4.3. Tools

The implementation in this research is mainly written in Python programming language, includes data preparation, model training, evaluation, and post-processing. The following sections provide a detailed description of the tools used in this research, including software, essential libraries and frameworks, and hardware.

4.3.1. Essential libraries and frameworks

- GDAL is an open-source geospatial library for reading, writing, and transforming raster and vector geospatial data formats.

- Rasterio is a Python library for reading and writing geospatial raster data, built on top of GDAL, used for GeoTIFF handling and spatial metadata access.
- Geopandas is a Python library extending Pandas for working with geospatial vector data, used for spatial operations, reprojections, and saving GeoPackage outputs.
- MapReader is a Python library for patch-based classification pipeline for large map collections, used for data loading, patch extraction, model training, inference, and evaluation, available at https://github.com/maps-as-data/MapReader.
- PyTorch is an open-source deep learning framework used to implement and train the CycleGAN image-to-image translation model and the Mask R-CNN instance segmentation model.
- Torchvision is a PyTorch companion library providing pre-trained models and utilities used here to load the Mask R-CNN architecture and apply non-maximum suppresssion.
- pycocotools is a Python library for COCO dataset evaluation, used here to compute MAP50 metrics during Mask R-CNN training.
- Building-Regulariser is a post-processing tool used to regularize the building footprints extracted from the instance segmentation model, available at https://github.com/DPIRD-DMA/Building-Regulariser.
- Weights and Biases (WandB) is a machine learning experiment tracking tool used to log and visualize the training process of the deep learning models, used here to monitor the training of both CycleGAN and Mask R-CNN models, available at https://wandb.ai/.
- AllMaps CLI is a command-line interface tool for programmatically accessing and manipulating the IIIF resources such as georeference annotations of the historical mapsheets, available at https://github.com/allmaps/allmaps/tree/develop/apps/cli. AllMaps XYZ Tiles API is also used to access the historical maps in the form of XYZ tiles for visualization and exploration purposes, available at https://observablehq.com/@allmaps/allmaps-tile-server.

4.3.2. Software

- QGIS used for spatial data inspection, visualization, and manual verification of results. Additional plugins are also used for specific tasks including PDOK plugins for accessing BGT data.
- GIMP is an open-source raster image editor used for visual inspection of the map images.

4.4. Hardware

All implementation is performed on a laptop computer with the following hardware specifications:

- CPU Intel Core with 14 physical cores (20 logical cores)
- GPU NVidia Geforce RTX 4050 with 6GB VRAM, CUDA 12.7
- RAM 16GB
- OS Windows 11

5. Result and Analysis

TODO: Instructions

- Method/Approach 1 described in Section 5.1 - Section 5.5
- Method/Approach 2 described in Section 5.6 - Section 5.9

5.1. Data preparation

5.1.1. Acquiring bonnebladen maps

TODO: instructions

- Downloading process using AllMapsCLI vs IIIF Presentation API
- Data details (e.g. number of sheets, temporal coverage, spatial coverage, etc.) - charts
- Temporal aspect, verzienden, herdruk,
- Vaienti2025, historical maps itu dicopy dari versi sebelumnya
- Secret military purposes (Geheim)

5.1.2. Assessment of bonnebladen maps

5.2. Prepare training dataset

5.3. Model training and evaluation

INitial training Fine-tune

5.4. Comparison with HGN-1900

5.5. Downloading Rotterdam municipality maps

Assessment of Rotterdam municipality maps

5.6. Vector style transfer

Table hexcode dengan each layer

5.7. Image-to-image translation

5.8. Instance segmentation

5.9. Change analysis

6. Conclusion

6.1. Conclusion

6.2. Discussion

6.3. Future Work

A Declaration of AI/LLM usage

If you have used Artificial Intelligence (AI), Large Language Models (LLMs)—such as Claude, ChatGPT, Gemini, Copilot, or similar tools—at any stage of your MSc thesis, please explicitly disclose this in your document. Include the following details:

1. **What:** Specify which AI/LLM tool(s) you used (eg ChatGPT, Claude, Copilot, etc.).
2. **How:** Briefly describe how you used these tools (eg generating outlines, writing, coding, data analysis, editing, etc.).
3. **Extent:** Indicate the extent of their use (eg occasional assistance, regular drafting, checking grammar, etc.).
4. **Ethics:** Confirm adherence to academic integrity and that AI/LLM did not generate falsified, fictional, or plagiarized content.

Example statement: "Some parts of this thesis benefited from the use of AI tools such as ChatGPT for initial text drafts and grammar suggestions. All results, analyses, and conclusions are my own, and I verified that the content is original and meets academic integrity standards."

B Reproducibility self-assessment

To promote open science and research integrity, please include a statement about the reproducibility of your MSc thesis work.

Address the following points concisely:

1. **Data Availability:** Is your data (that you used as input, or that you generated) publicly accessible? If yes, provide a persistent link (ideally a DOI, you can archive our datasets on GitHub or in the 4TU repository); if no, briefly explain why not (eg privacy, licensing).
2. **Code Availability:** Is your code and analysis workflow openly available? This can be for instance via GitHub, OSF, Zenodo. Include the relevant link or DOI if applicable. It is suggested to follow those guidelines for reproducibility: https://geomatics.bk.tudelft.nl/geogeeks/git/goodgit/.
3. **Documentation:** Have you provided clear instructions, README files, or notebooks so others can understand and reproduce your results?
4. **Reproducibility Rating:** Self-rate your overall reproducibility using the following scale:
   - **High:** All data and code fully available, well-documented, and tested to reproduce main results.
   - **Moderate:** Most data and code available, basic documentation, some steps may require clarification.
   - **Low:** Limited or no data/code sharing, minimal documentation, reproduction unlikely.

B.1 Example statement

"All data and analysis code related to this thesis are openly accessible at `[insert link]`. Complete instructions and a sample workflow are provided in the project README. I rate the reproducibility of this thesis as **high** according to the provided scale."

B.2 Examples of good MSc Geomatics repositories

1. https://github.com/chenzhaiyu/points2poly
2. https://github.com/ellieroy/no-floors-inference-NL
3. https://github.com/NoortjevanderHorst/treegrowthmodelling
4. https://github.com/fabisser/stylesdf

C Some UML diagrams

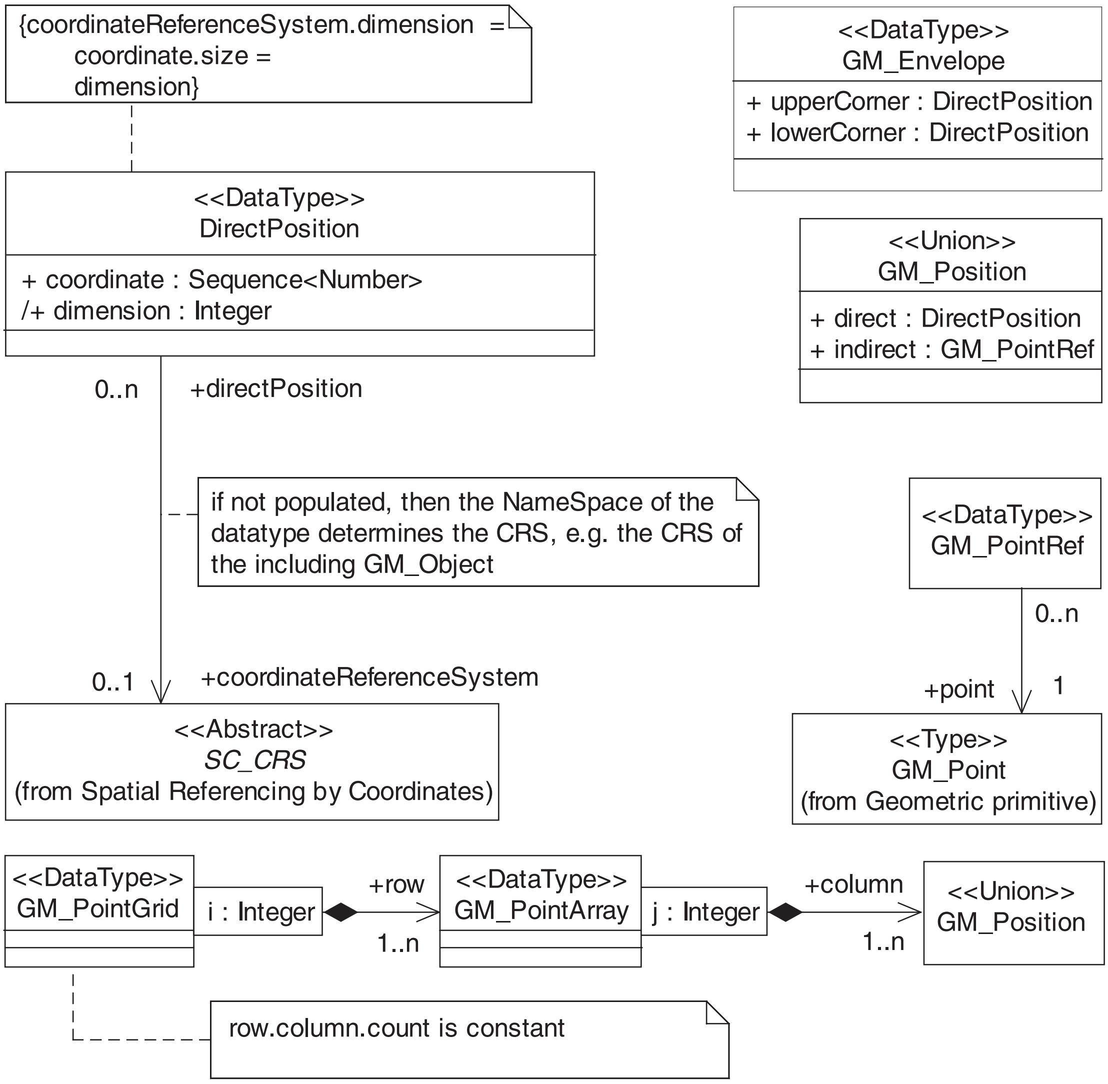

Figure C.1: The UML diagram of something that looks important.

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